HMITiles
An Open-Source HMI Tile Library for Industrial Dashboards

Overview
HMITiles is an open-source HMI (Human Machine Interface) tile library written in **B4X**, following widely accepted industrial HMI principles.

It provides reusable, professional-grade HMI tiles for: Industrial dashboards, SCADA (Supervisory Control and Data Acquisition) front-ends, Machine and process HMIs.

The focus of this library is **clarity, consistency, and disciplined HMI design -** not visual effects or UI decoration.

This is a **non-commercial project** shared for inspiration.

B4J Example – Overview HMITiles

HMITiles Used
HMITileLabel (3)
HMITileDigitalClock
HMITileClock
HMITileButton (3)

HMITileSensor (3)
HMITileSeekBar (2)
HMITileGauge

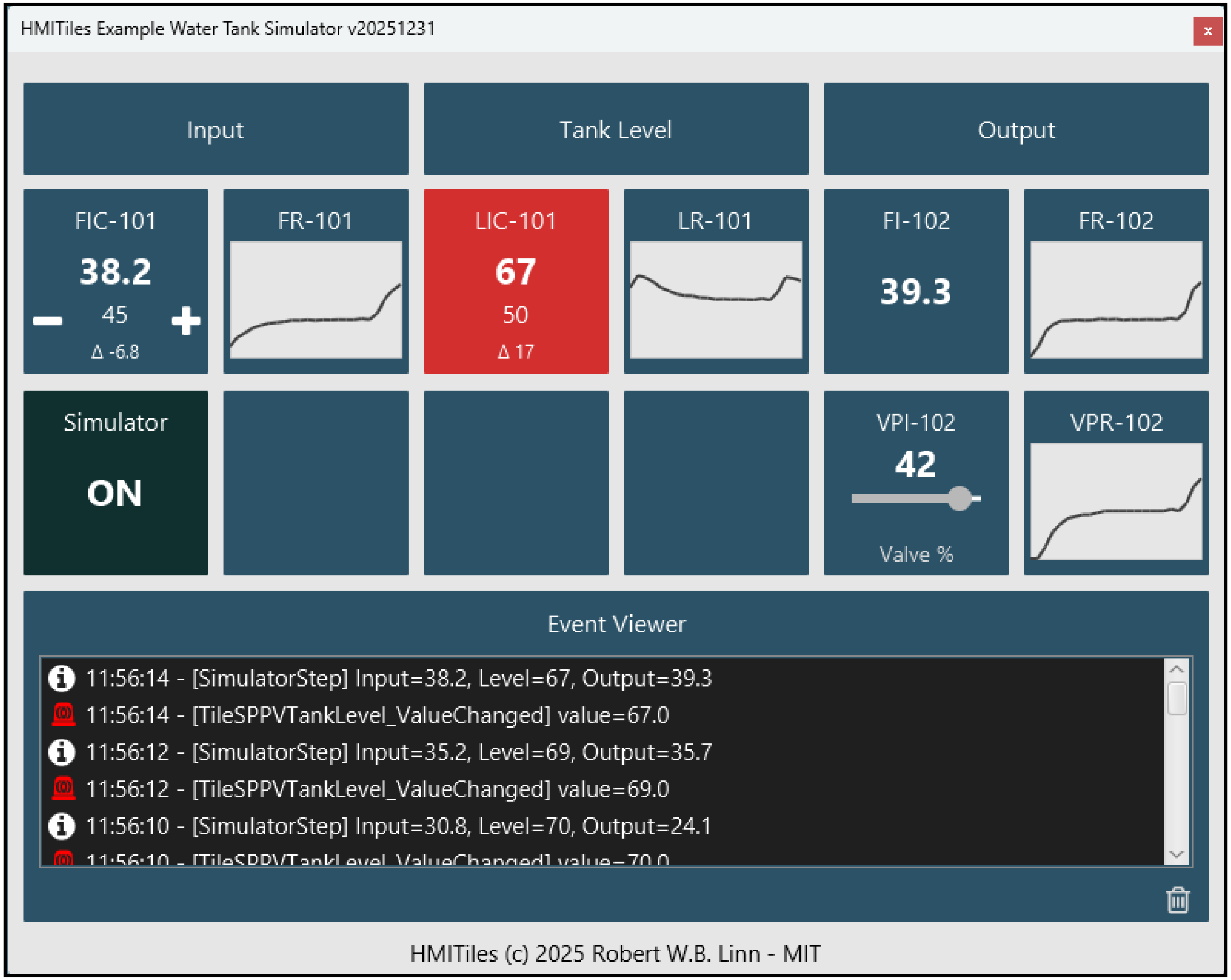
HMITileRGB (2)
HMITileLevel
HMITileReadOut
HMITileTrend
HMITileImage

HMITileSetPoint
HMITileSelectList
HMITileSelect
HMITileShape (8)
HMITileImageIcon

HMITileEventViewer
HMITileList

B4J Example - Water Tank Simulator

A simple interactive water tank PID simulator demonstrating selective HMITiles.



HMITiles Used

HMITileLabel

HMITileSetpoint
HMITileTrend
HMITileSetpoint
HMITileReadOut
HMITileTrend

HMITileButton
HMITileSeekBar
HMITileTrend

HMITileEventViewer

B4J Example – ArduinoLED

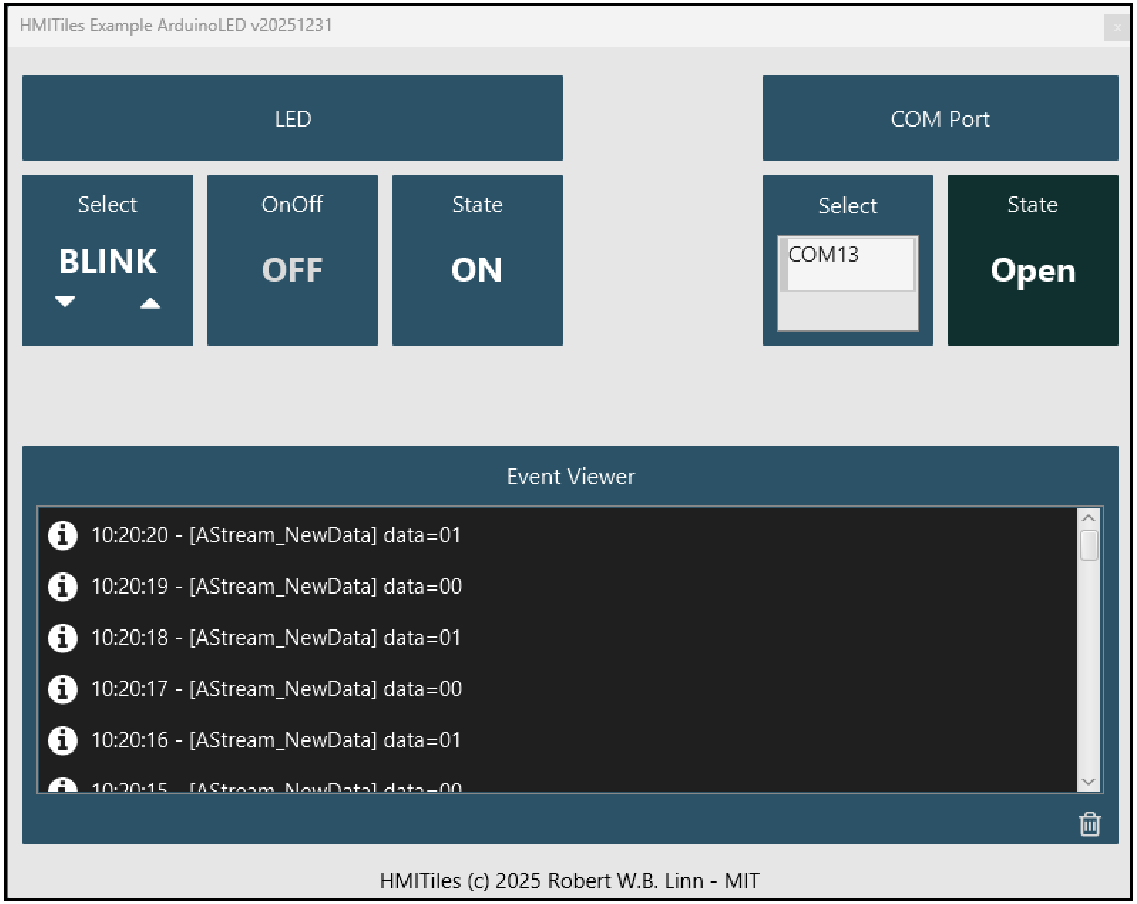
Control the state of an LED connected to an Arduino UNO via the serial line.

HMITiles Used

HMITileLabel

HMITileSelect
HMITileButton
HMITileReadOut

HMITileEventViewer



HMITiles Used

HMITileLabel

HMITileSelectList
HMITileButton

HMITileEventViewer

B4A Example – Overview

HMITiles Used

HMITileLabel (3)
HMITileDigitalClock
HMITileClock

HMITileButton (3)
HMITileTrend
HMITileGauge

HMITileSensor
HMITileReadOut
HMITileLevel

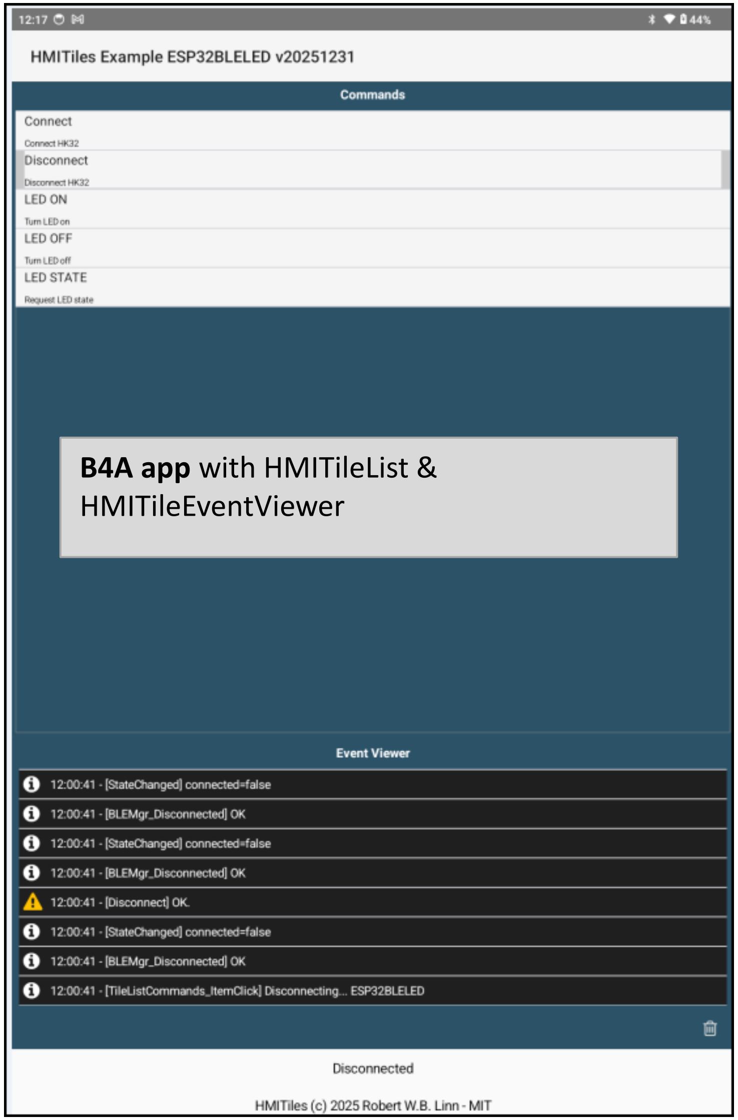
HMITileSeekBar (2)

HMITileSetPoint
HMITileSelectList
HMITileSelect

HMITileEventViewer

B4A Example – ESP32BLELED

Control the state of an LED connected to an ESP32 via the BLE (UART).



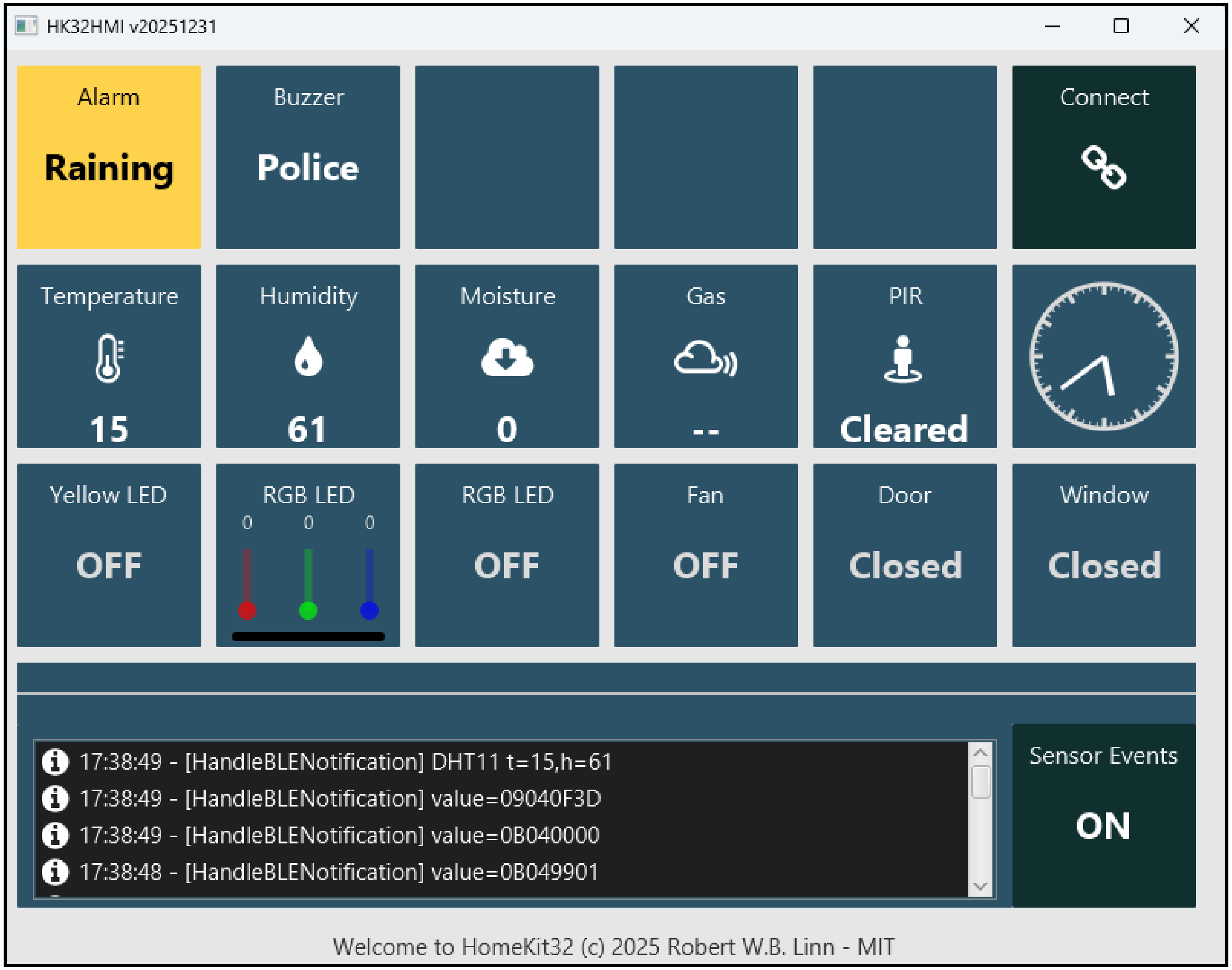
B4A app with HMITileList & HMITileEventViewer

```
[B4RBLEServer::onConnect] Client connected
[CommBLE.BLEServer_NewData] buffer HEX=010101
[CommBLE.BLEDispatch] deviceid=1
[DevLED.ProcessBLE] payload=010101
[CommBLE.BLEServer_Write] data=010101
[B4RBLEServer::Write] Notified bytes: 3
[DevLED.WriteToBLE] payload=010101
[B4RBLEServer::onWrite] Received bytes: 3
[CommBLE.BLEServer_NewData] buffer HEX=010100
[CommBLE.BLEDispatch] deviceid=1
[DevLED.ProcessBLE] payload=010100
[CommBLE.BLEServer_Write] data=010100
[B4RBLEServer::Write] Notified bytes: 3
[DevLED.WriteToBLE] payload=010100
[B4RBLEServer::onWrite] Received bytes: 3
[CommBLE.BLEServer_NewData] buffer HEX=0102
[CommBLE.BLEDispatch] deviceid=1
[DevLED.ProcessBLE] payload=0102
[CommBLE.BLEServer_Write] data=010200
[B4RBLEServer::Write] Notified bytes: 3
[DevLED.WriteToBLE] payload=010200
[B4RBLEServer::onWrite] Received bytes: 2
[B4RBLEServer::SetStartAdvertising] Started advertising: ESP32BLELED
[B4RBLEServer::onDisconnect] Client disconnected, restarting advertising
```

B4R program IDE log showing the commands:
connect > command LED ON > command LED OFF > disconnect

B4J Example – HomeKit32

HomeKit32 is a modular smart-home automation framework for the Keyestudio Smart Home Kit (KS5009), powered by **ESP32, BLE, MQTT and B4X**.



HMITiles Used

HMITileLabel (2)
HMITileButton

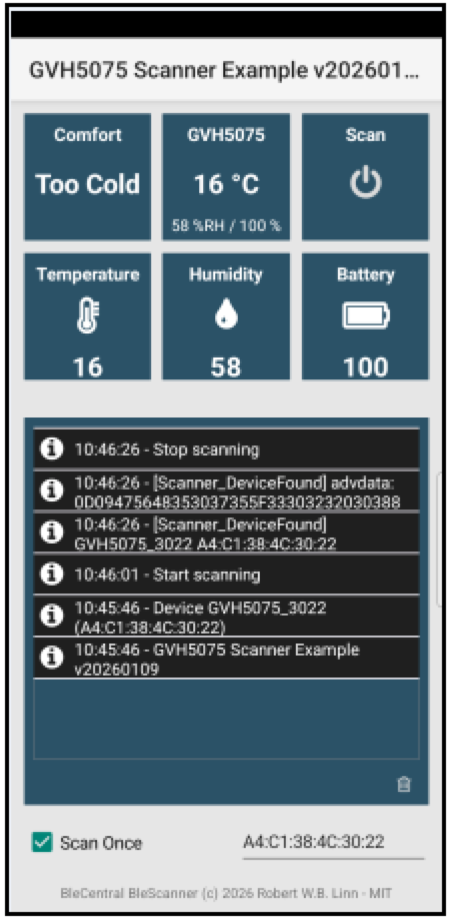
HMITileSensor (4)
HMITileDigitalClock
HMITileClock

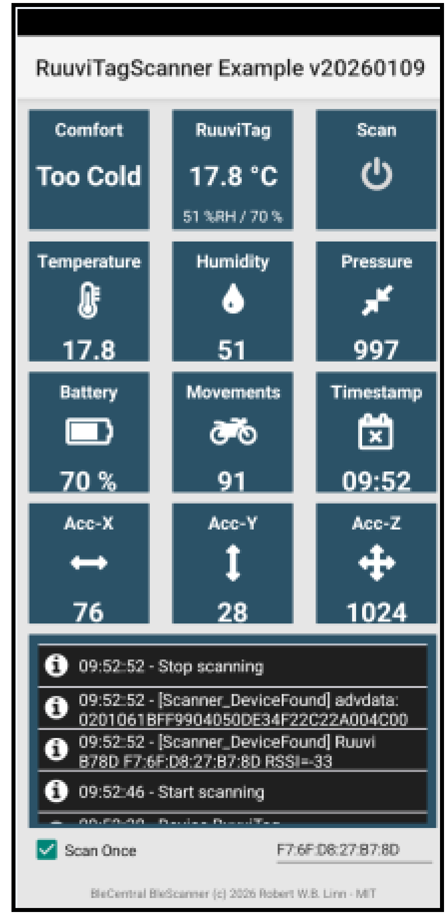
HMITileButton (5)
HMITileRGB

HMITileEventViewer
HMITileButton

B4A Examples – BLE GVH5075 & RuuviTag

B4A additional library **BleCentral** examples scanning devices Govee GVH5075 & RuuviTag.





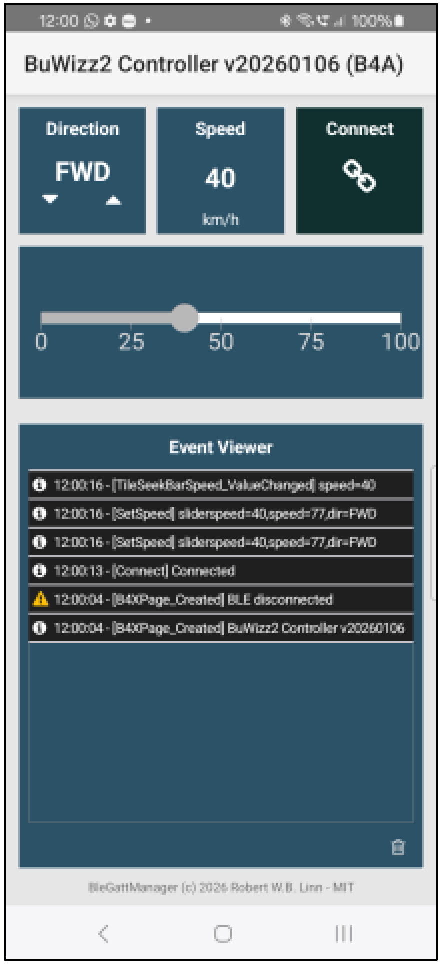
HMITiles Used

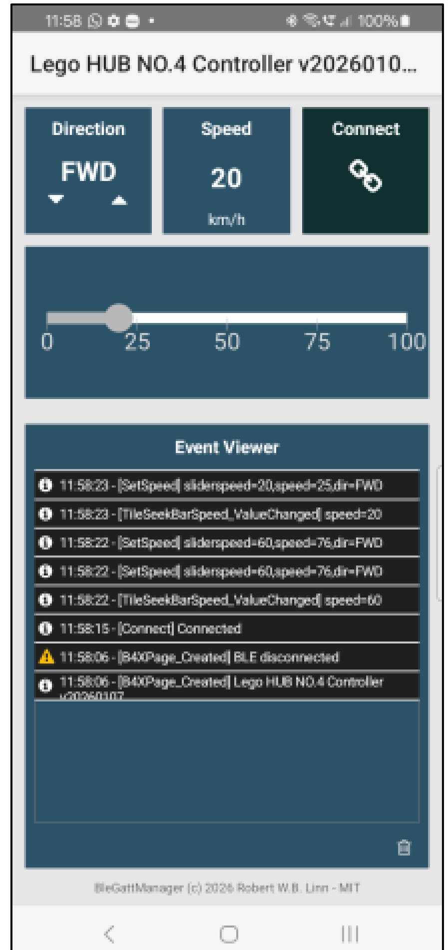
HMITileButton
HMITileSensor

HMITileEventViewer

B4A Examples – BLE BuWizz2 & Lego HUB No.4

B4A additional library **BleCentral** examples controlling bricks BuWizz2 & Lego HUB No.4.





HMITiles Used

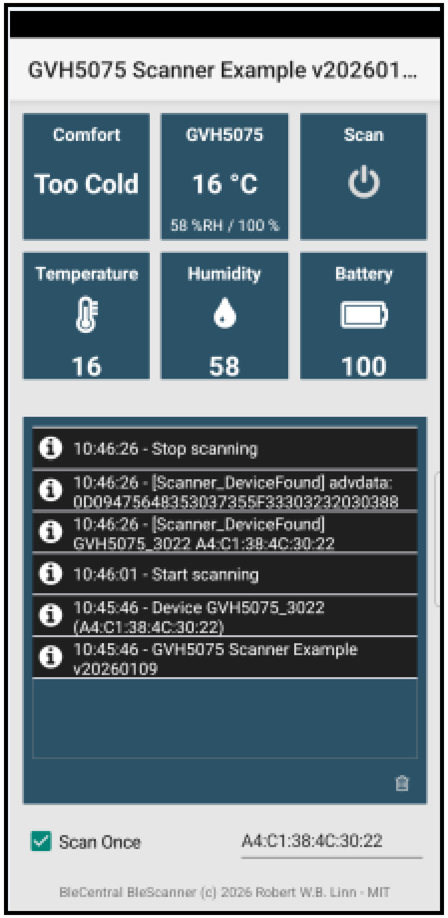
HMITileSelect
HMITileReadOut
HMITileButton

HMITileSeekBar

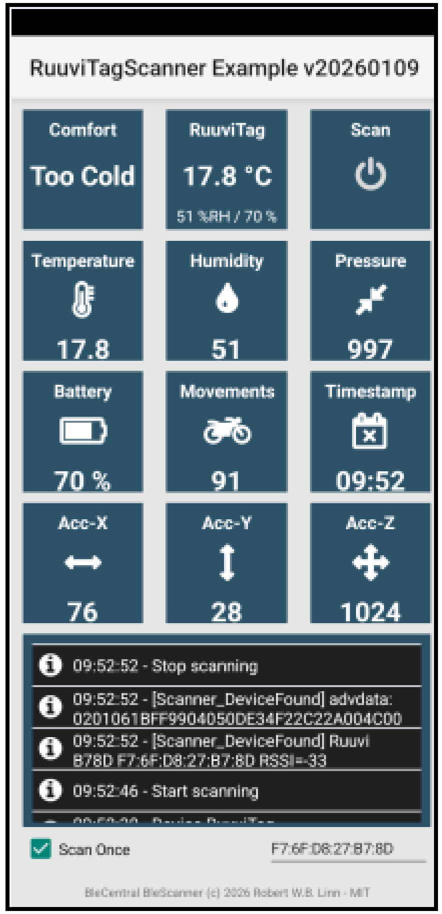
HMITileEventViewer

B4A Examples – BleCentral.b4xlib

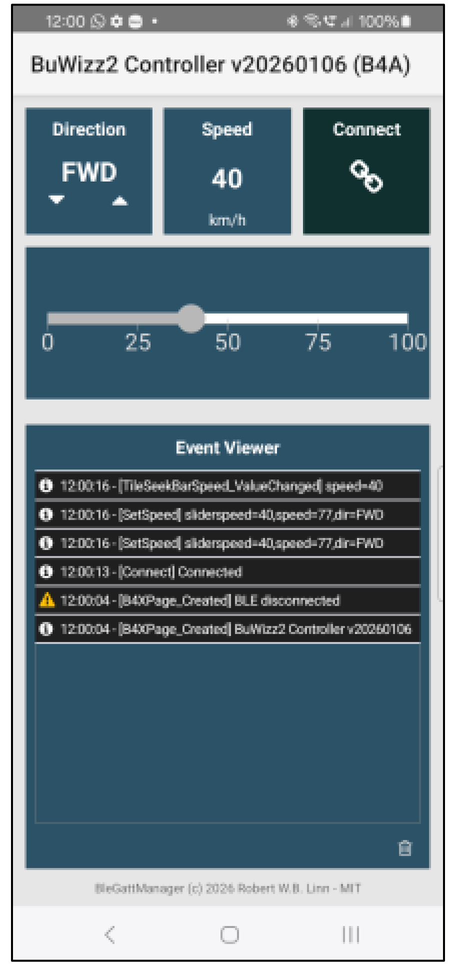
HMITiles **B4A** examples using the B4A library **BleCentral**.b4xlib



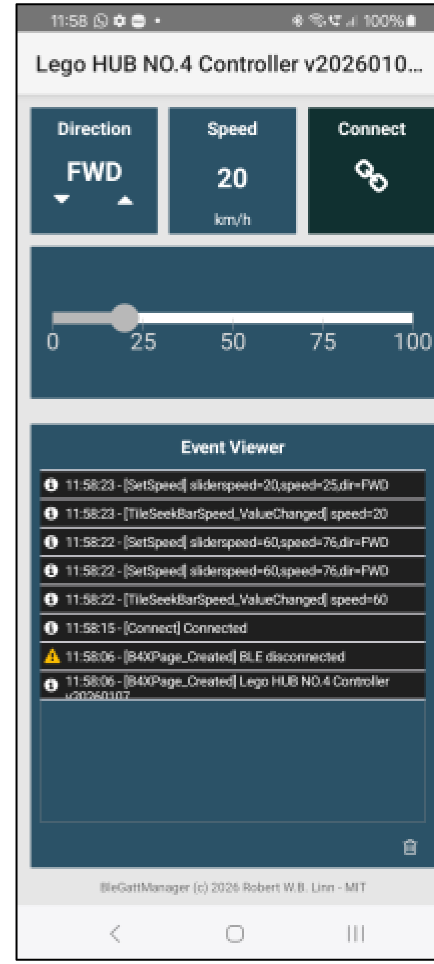
Govee Bluetooth hygrometer and thermometer.



Open-source Bluetooth sensor temperature, humidity, air pressure, and motion.



Compact, rechargeable battery and Bluetooth remote control brick for Lego models.



Lego Powered Up Bluetooth Hub.

Screenshots